

The objective of this assignment is to design and implement a basic Python application that interacts with a database system, performing CRUD (Create, Read, Update, Delete) operations.

Assignment Description: In this assignment, each student is required to develop a Python application that allows for the manipulation of data stored in a database. The application should be able to perform the following CRUD operations:

1. **Create (Insert):** Insert new records into the database.
2. **Read:** Retrieve and display records from the database.
3. **Update:** Modify existing records in the database.
4. **Delete:** Remove records from the database.

You need to write separate functions for each of these operations as shown in the recitation session. They do not need to be general purpose functions; therefore, you can tailor them based on your tables' needs (like the basic example in the recitation). However, you shouldn't directly execute an SQL query, but you need to inject the inputs you provide into the queries, where you must take the basic precautions shown in recitation against SQL Injection Attack.

You will only use the two tables you have chosen for the previous project phase. You must carry out all the operations that are mentioned above for each table. Before and after executing the functions, you need to take a printout of the whole table to show the change in the data. You must add these logs to the **END** of your report. If your table contains too many items to show, then you can limit the number of records you return by adding a (limit 10) to only show 10 records to be retrieved, where the important record that has been affected is included in these 10 records.

You will submit a zip file containing the code you have written and a project report. The project report will contain:

1. A brief description about what you have done in this step,
2. The tables you have chosen and the data they contain.
3. The operations, queries you generated using your functions.
 - a. It would be better for you to print your generated queries before executing for this.
4. The screenshots of before and after logs mentioned above.

Your project files must contain these sections, and they must be in this order, otherwise you may lose points.

This assignment is **an individual assignment**; therefore, you will submit your code and reports individually as well.

You may contact the assistants for your questions regarding to the project step:

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